# Assessments

**MATH 107: Introduction to Linear Algebra — Fall 2026**

# Labs

Lab points (effort) are earned in class: work the hub exercises in Octave or Python, handwrite answers on the one-pager hard copy, and get check-in initials after a short conversation. Exercises are on each lab page in both languages; the one-pager is linked from the hub so you can preview it. Quizzes may ask about lab ideas. See Lab 0 for the pattern.

| Lab | Date | Learning objectives |
|---|---|---|
| Lab 0: Computers Speak Vectors | Fri Aug 28 | Enter, index, and modify vectors in Octave or Python; plot and debug supplied code. |
| Lab 1: Finding Similar Things | Fri Oct 16 | Compute distances; name a nearest neighbor; see the effect of a bad feature scale. |
| Lab 2: Gaussian elimination | Fri Oct 23 | Eliminate with elementary matrices; back-substitute by hand; solve and check $Ax = b$ in Octave or Python. |
| Lab 3: Mixing and Balancing | Fri Oct 30 | Write a small system $Ax = b$; solve in software; check the residual; prefer a solver over an explicit inverse. |
| Lab 4: Predict Through Time | Fri Nov 6 | Implement $x_{t+1} = Ax_t$; iterate; hand-check early steps; describe a long-run pattern. |
| Lab 5: Steady State and Eigen | Fri Nov 20 | Find a steady state / $\lambda = 1$ direction; match eigenpairs to Lab 4's long-run mix. |
| Lab 6: Fit and Compress | Fri Dec 4 | Least-squares fit; truncated SVD; residual and storage $k(m + n + 1)$ — warm-up for the capstone. |

Labs 1–6 (and quizzes) meet on **Fridays**. There is no lab on Midterm 2 day or during fall break. Capstone work uses the capstone packet, not a separate Lab 7.

# Quizzes

Eleven quizzes. Quiz 7 (Fri Oct 23) is online in Canvas. The others are in the first 10 minutes of class. The lowest two scores are dropped. Practice packets are grouped as **Core** and **Stretch**. Each Friday quiz covers **only the prior instructional week** (not that day's new lecture or lab).

A consistent quiz has three parts:

- **Fluency** (4–5 points): one small calculation closely modeled on a practice item.
- **Meaning** (3–4 points): explain or interpret what the calculation says.
- **Computation/application** (2–3 points): interpret code, a plot, a residual, or software output.

You will not write substantial code from scratch on a timed quiz. Typical prompts: complete one missing line, predict three lines of output, identify an error, interpret output, or choose an appropriate command.

Every second or third quiz includes a short cumulative item from an earlier packet (beginning with Quiz 3).

## Source key

| Code | Source |
|---|---|
| B 3.10 | Boyd and Vandenberghe, Exercise 3.10 |
| S 6.1 #10 | Strang, Section 6.1, Problem 10 |
| L Ch. 2 Ex. 3 | Lachniet, *Introduction to GNU Octave* (3rd ed., 2020), Chapter 2, Exercise 3 |
| C | Course-created bridge or lab checkpoint |

A "prompt" can be one selected part of a multi-part textbook exercise.

## CLO coverage

Official course learning outcomes are assessed more than once; nothing important is a single-day checklist item.

| Required outcome | Where it is developed and assessed |
|---|---|
| Interpret vectors, matrices, and complex numbers algebraically and geometrically | Weeks 1–3, 7, and 13 |
| Solve systems by row reduction by hand and with software | Lectures 21–22 and Lab 2; Midterm 2; cumulative final |
| Find $2 \times 2$ eigenpairs by hand and general eigenpairs with software | November 18 and Lab 5 |
| Apply eigenvalues and eigenvectors | Dynamics in Week 9, explained in Week 13; Markov/normal-mode applications |
| Recognize triangular, symmetric, and positive-definite matrices | October 5, November 6, and December 2 |
| Use linear combination, subspace, span, independence, basis, and dimension | Weeks 1 and 5, revisited in Weeks 8 and 11 |

| Required outcome | Where it is developed and assessed |
|---|---|
| Write or adapt programs involving matrices, vectors, and plotting | Labs 0–6 and the capstone, with steadily reduced scaffolding |
| Apply linear algebra to real-world problems | Every lab and most class examples |
| Learn independently and collaboratively | Practice banks, low-stakes quizzes, paired work, lab studio, and capstone presentation |
| ILO/GEAR quantitative reasoning in context | Interpretation questions throughout; least-squares and capstone conclusions |

## Quiz dates

Friday quizzes cover **only the prior instructional week** (skip midterm week and fall break when choosing that week). Same-day lecture/lab topics are not on that day's quiz.

| Quiz | Date | Focus (prior week) |
|---|---|---|
| Quiz 1 | Fri Sep 4 | Vectors, arithmetic, Lab 0 (Week 1) |
| Quiz 2 | Fri Sep 11 | Combinations, inner product, linear functions (Week 2) |
| Quiz 3 | Fri Sep 18 | Applications and norms (Week 3) |
| Quiz 4 | Fri Sep 25 | Angle, nearest neighbors, and clustering (Week 4) |
| Quiz 5 | Fri Oct 9 | Dependence, span, and basis (Week 5) |
| Quiz 6 | Fri Oct 16 | Matrices and $Ax$ (Week 7 / Lectures 18–19) |
| Quiz 7 | Fri Oct 23 | Transforms only (Lecture 20). Online in Canvas. |
| Quiz 8 | Fri Oct 30 | Systems, elimination, RREF, dynamics, and iteration (Lectures 21–23) |
| Quiz 9 | Fri Nov 6 | Products, powers, and cost (Lectures 24–26) |
| Quiz 10 | Fri Nov 20 | Inverses, rank, and determinants (Lectures 27–29) |
| Quiz 11 | Fri Dec 4 | Eigenvalues, complex numbers, and Lab 5 (Week 13 / Lectures 35–37) |

**Not on a dedicated quiz** (assessed on Midterm 2, the final, and/or the capstone): same-day lecture/lab topics on quiz Fridays; Week 15 least squares / SVD. Use the Midterm 2 practice packet, final practice packet, and capstone packet as appropriate.

# Exams

In-class, for the entire class period.

| Exam | Date | Learning objectives |
|---|---|---|
| Midterm 1 (in class; not posted) | Fri Oct 2 | Compute with vectors and linear functions; interpret operations; analyze distance, similarity, span, and independence; select and communicate a method. Day page: Lecture 16. Prep: study guide, practice. |
| Midterm 2 | Fri Nov 13 | Analyze matrices as transformations and models; solve and interpret systems; use products, powers, rank, inverses, and determinants; interpret computational results. Prep: study guide, practice. |
| Final exam | Wed Dec 16, 10:20 a.m.–12:10 p.m., FR 201 | Hand fluency; select tools in new contexts; interpret software; explain connections among major concepts; apply linear algebra to quantitative problems. Prep: study guide, practice. |

# Project

| Project | Dates | Learning objectives |
|---|---|---|
| Capstone | Dec 4 launch, Dec 7 studio, Dec 9 gallery | Low-rank SVD strategy; compare reconstruction error, storage, and runtime; present quantitative evidence and limitations. |

# Class participation

Attendance and participation in class.